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Editors

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Re: Support for the proposed book on high-speed electrical machines and drives

Dear Editors,

I am pleased to support the proposal for the book on high-speed electrical machines and drives and to confirm my intention, subject to the publishing contract, to contribute the chapter on thermal management and cooling architectures.

My research has focused for more than 25 years on the thermal management of electrical machines, including winding heat-transfer paths and thermal resistances, material and interface effects, indirect and direct cooling architectures, and advanced approaches such as direct conductor cooling and additive-manufactured cooling structures. From this perspective, I recognise the design problem around which the proposed book is structured.

Thermal management is still frequently treated as a downstream activity in electrical-machine design: the electromagnetic design establishes the losses and the thermal analysis then determines whether these losses can be removed while maintaining acceptable component temperatures. For high-speed and high-power-density machines, this sequential approach becomes increasingly inadequate. The cooling architecture itself influences the admissible loss density, electromagnetic loading, component temperatures and, ultimately, the achievable machine size and performance. Thermal management therefore needs to be considered as an integral part of the multidisciplinary design process alongside electromagnetic, mechanical and rotordynamic constraints.

There is a substantial body of literature concerned with thermal analysis of electrical machines, including analytical, lumped-parameter and numerical approaches for predicting the temperatures of an already defined design. Comparatively less attention has been given to thermal architecture selection as a design problem: for example, how to choose between conventional housing or water-jacket cooling, in-slot cooling and direct winding or conductor cooling, and how these alternatives influence achievable loss density, packaging, manufacturability, reliability and overall machine performance. In my view, treating these cooling architectures explicitly as design choices would be a valuable contribution of the proposed book.

The rotor presents a further important challenge in high-speed permanent-magnet machines. Heat generated within the rotor must be removed through thermal paths strongly influenced by the mechanical design, including the retaining sleeve and associated interfaces. At the same time, rotor and air-gap losses increase with operating speed and electrical frequency. Magnet temperature and demagnetisation margin can therefore become major design constraints and need to be considered together with electromagnetic performance and rotor mechanical integrity.

Subject to the publishing contract, I intend to contribute the chapter on thermal management and cooling architectures. The chapter would address winding heat paths and thermal resistances; stator cooling architectures ranging from conventional water jackets to in-slot and direct winding or conductor cooling; rotor cooling and the influence of retaining structures; air-gap heat transfer; thermal-network modelling and calibration; transient thermal behaviour under realistic duty cycles; and experimental measurement and validation. My intention would be to present these topics not simply as individual cooling technologies, but as elements of an integrated thermal design methodology for high-speed electrical machines.

Having reviewed the proposed structure and introductory material, I believe the book addresses an important and timely engineering problem. Its multidisciplinary treatment of high-speed electrical-machine design, in which mechanical, electromagnetic and thermal constraints must be balanced simultaneously to obtain a feasible design for a particular application, should make it a useful contribution to the electrical-machines community.

I am therefore pleased to support the proposal and to participate as the author of the thermal management and cooling architectures chapter.

Yours sincerely,

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